Analytics Skills Deep Dive

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> **What are Analytics Skills?**
> Skills are pre-built knowledge packages that tell the AI assistant exactly which tables, columns, filters, and query patterns to use for each business domain. Instead of users needing to know the data schema, they just ask questions in plain English — the skill handles the rest.

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How Skills Work

The Problem

Groww's data platform has **hundreds of tables** across multiple schemas, with billions of rows, inconsistent naming, and complex join relationships. Even experienced analysts spend significant time finding the right table, understanding column semantics, and writing efficient queries.

The Solution

Each skill encodes deep domain knowledge:

What the Skill Knows	Example
Which tables to use	"For stock orders, use `core_stocks_order`, NOT the stale `stocks_order`"
Partition columns	"Always filter on `order_date` — this table has 2.5B rows"
Column semantics	"`product = 'CNC'` means delivery, `'MIS'` means intraday"
Filter values	"Order status values: `EXECUTED`, `CANCELLED`, `REJECTED`, `FAILED`"
Query patterns	Pre-written SQL templates for common questions
Data quality warnings	"This table has future-dated rows — always cap your date range"

Join keys	"Join on `user_account_id` — it's the universal key"
Stale table warnings	"DO NOT USE `invest_user_fid_info` — stale since Jun 2024"

User Experience

```
User:  "What was the stock order success rate yesterday?"

Without Skills:
  - User needs to know `core_stocks_order` is the right table
  - User needs to know `order_date` is the partition column
  - User needs to know `EXECUTED` means success
  - User writes SQL manually

With Skills:
  - AI loads the Stocks & Trading skill
  - Writes: SELECT order_date, COUNT(*) FILTER (WHERE order_status = 'EXECUTED') ...
  - Returns: "Yesterday's success rate was 87.3% across 14.2M orders"
```

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1. Stocks & Trading Analytics

> **Trigger**: Stock orders, equities, delivery/intraday, turnover, brokerage, CNC/MIS, AMO, GTT, SIP orders, ETF, platform mix

Tables

Table	Schema	Rows	Grain	Partition
`core_stocks_order`	`core_invest`	2.5B	Per order	`order_date`
`fact_transaction_stocks`	`core_invest`	7.1B	Per trade/leg	`order_date`
`order_level_revenue_with_order_type`	`core_invest`	2.5B	Per order + revenue	`trade_date`

Key Dimensions

Dimension	Column	Values
Product	`product`	`CNC` (delivery), `MIS` (intraday), `MTF` (margin)
Direction	`buy_sell`	`BUY`, `SELL`
Status	`order_status`	`EXECUTED`, `CANCELLED`, `REJECTED`, `FAILED`, `NEW`, `OPEN`
Exchange	`exchange`	`NSE`, `BSE`
Platform	`order_platform`	`android`, `ios`, `web`
Order Type	`order_type`	`LIMIT`, `MARKET`, `SL`, `SL-M`
Special Flags	`is_amo`, `is_gtt`, `is_sip_order`, `is_etf_order`	`true` / `false`

Example Questions It Can Answer

- “How many stock orders were executed yesterday?”
- “What’s the CNC vs MIS turnover split this week?”
- “Top 20 stocks by trading volume”
- “AMO and GTT adoption trends”
- “Brokerage revenue by product type”
- “Platform mix — app vs web”
- “Stock SIP order growth”
- “Order success rate by hour of day”

Key Guardrails

- Always filter on `order_date` partition — 2.5B rows
- Prefer `core_stocks_order` over stale `stocks_order` (last updated Oct 2025)
- Use `fact_transaction_stocks` only when trade-level granularity needed (multi-fill, lifecycle timestamps)
- Never query `fact_transaction_stocks` (7.1B rows) for more than 7 days without aggregation

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2. F&O (Futures & Options) Analytics

> **Trigger**: F&O orders, options/futures, PnL, ITM/ATM/OTM, power traders, basket orders, fast exit, strike price, expiry, derivatives

Tablesf

Table	Schema	Rows	Grain	Partition
`fno_order_master`	`core_invest`	3.8B	Per order	`pt`
`fno_pnl_master`	`core_invest`	1.9B	Per position exit	`pt`
`fno_txn_master`	`core_invest`	557M	Per enriched txn	`pt`
`fno_usersegmentation_monthly`	`core_invest`	138M	Per user/month	`month`
`fno_livemonitoring`	`core_invest`	~133	Per 5-min interval	Real-time
`fno_softblockuserstatus_list`	`core_invest`	32	Per blocked user	Snapshot
`fno_fastexit_v1_dailytxn_v1`	`core_invest`	15.5K	Per day/segment	Daily
`safeexit_basetable`	`core_invest`	720K	Per safe exit trigger	Daily
`fact_onboarding_fno`	`core_invest`	48M	Per user F&O journey	`fnostart_`

Key Dimensions

Dimension	Column	Values
Contract Type	`contract_type`	`FUTURES`, `OPTIONS`

Call/Put	`call_put`	`CE` (call), `PE` (put)
Moneyness	`itm_atm_otm`	`ITM`, `ATM`, `OTM`
Underlying	`contracts_underlying`	`NIFTY`, `BANKNIFTY`, `FINNIFTY`, stock names
Power Trader	`power_trader`	`true` / `false`
Basket Order	`is_basket_order`	`true` / `false`
Fast Exit	`is_fast_exit`	`true` / `false`
API Order	`api_flag`	`true` / `false`

Example Questions It Can Answer

- “F&O order volume and success rate”
- “Options vs Futures breakdown”
- “ITM/ATM/OTM distribution”
- “Power trader vs regular trader comparison”
- “User PnL distribution”
- “Expiry day trading analysis”
- “Basket order and fast exit adoption”
- “F&O live health check — is success rate normal?”
- “Who are the soft-blocked users?”
- “Safe exit (loss limit) trigger effectiveness”
- “F&O onboarding funnel with TATs”

Key Guardrails

- `fno_order_master.pt` contains future dates (up to 2090) — contract expiry dates leak in. Always add `pt < date('2027-01-01')` or filter `is_tradingday = true`
- PnL: `sell_value - buy_value` for long positions; sign reverses for shorts
- `fno_usersegmentation_monthly` stops at Oct 2025 — check freshness
- `fno_livemonitoring` retains only today’s data (~133 rows) — ephemeral

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3. Mutual Fund (MF) Analytics

> **Trigger**: Mutual fund orders, SIPs, NAV, AUM, folios, scheme performance, MF user metrics, MF engagement, churn, payment analysis

Tables

Table	Schema	Purpose
`dim_mutual_fund_scheme`	`dwh_invest`	Scheme master: NAV, ISIN, category, AMC, fund house
`fact_mutual_fund_order_details_enriched`	`dwh_invest`	Enriched order details with payment + FID flags
`fact_mutual_fund_sip_details`	`dwh_invest`	SIP lifecycle: mandates, step-ups, installments

`view_mf_user_aum_latest`	`dwh_invest`	Latest AUM snapshot per user/scheme/folio
`agg_daily_mf_user_aum`	`dwh_invest`	Daily AUM trends
`snap_daily_mf_scheme_nav`	`dwh_invest`	Daily NAV snapshots
`mf_user_properties`	`dwh_invest`	User-level lifetime metrics
`agg_mf_user_engagement_metrics`	`dwh_invest`	Engagement funnels (D/W/M grain)
`agg_*_orders_transactional_metrics`	`dwh_invest`	Pre-aggregated order metrics (D/W/M)
`agg_*_users_transactional_metrics`	`dwh_invest`	Pre-aggregated user metrics (D/W/M)

Table Selection Strategy

Analysis Type	Use This Table
Order volumes, success rates, payments	`fact_mutual_fund_order_details_enriched` or `agg_*_orders_*`
SIP creation, mandates, cancellations	`fact_mutual_fund_sip_details`
Current AUM / portfolio	`view_mf_user_aum_latest`
AUM trends over time	`agg_daily_mf_user_aum`
NAV / scheme performance	`dim_mutual_fund_scheme` + `snap_daily_mf_scheme_nav`
User segmentation / LTV	`mf_user_properties`
Engagement funnels	`agg_mf_user_engagement_metrics`
Scheme lookup	`dim_mutual_fund_scheme`

Example Questions It Can Answer

- “How many MF orders were placed this week?”
- “SIP creation and cancellation trends”
- “Top schemes by AUM”
- “MF order completion rate by payment type”
- “User lifetime invested vs redeemed”
- “Scheme category breakdown (equity, debt, hybrid)”
- “MF user engagement funnel (landing → payment → completion)”
- “Active vs inactive folio analysis”

4. User Master & Growth Analytics

> **Trigger**: User profiles, signup funnels, FID tracking, AUM, segmentation, referrals, uninstalls, cohort analysis, demographics

Tables

Table	Schema	Rows	Purpose
`growth_user_master`	`core_bgv`	117M	Core user dimension: demographics, KYC, signup
`growth_user_master_ultimate`	`core_bgv`	117M	Enriched 360 view: AUM, revenue, device, quality flags
`growth_realtime_productwise_fids`	`core_bgv`	89M	FID per product (deduplicated)
`growth_realtime_productwise_fids_streaming_v3`	`core_bgv`	393M	Streaming FID with transaction numbers
`user_product_fid_fact`	`core_bgv`	156M	FID fact with placed vs executed split
`user_product_txn_master_fact`	`core_bgv`	**8.3B**	All orders across all products
`growth_user_peak_current_aum`	`core_bgv`	30M	Latest and peak AUM
`growth_user_power_trader`	`core_bgv`	11M	Monthly power-trader classification
`growth_user_referral`	`core_bgv`	7.6M	Referral chains and acquisition costs
`growth_user_uninstalls`	`core_bgv`	113M	App uninstall events

Key Dimensions

Dimension	Values
User Status	`SignUp` (72M), `Onboarded & Transacted` (27M), `Onboarded & Non-Transacted` (17M)
Source	`performance` (63M), `organic++` (47M), `referral` (7.6M)
Platform	`android` (98M), `ios` (7M), `msite` (3.7M), `desktop` (3.3M)
Age Group	`18-22`, `23-27`, `28-35`, `36-40`, `40+`
AUM Band	`Zero AUM`, `<1k`, `1k-10k`, `10k-50k`, `50k-1L`, `1L-5L`, `5L-10L`, `10L-50L`, `>=50L`
FID Product	`cnc`, `sip`, `lumpsum`, `mis`, `fno`, `gold`, `mtf`, `commodity`, ...

Example Questions It Can Answer

- “How many users signed up this month?”
- “Signup → onboarding → FID conversion funnel”
- “FID product mix trend (what are users investing in first?)”

- “D7 NTU (new-to-transact) rate by cohort”
- “AUM distribution by tier city”
- “Revenue per user by signup source”
- “Referral effectiveness — who brings the best users?”
- “Uninstall rate by cohort and device”
- “Power trader classification”

Key Guardrails

- `user_product_txn_master_fact` is **8.3B rows** — MUST filter on `order_date` partition
- Product type naming is **inconsistent** across tables (FID: `cnc`, `sip`; txn_master: `Stocks`, `Mf`)
- Universal join key: `user_account_id` (format `ACC<digits>`)
- Prefer `growth_user_master_ultimate` over `growth_user_master` for analytics

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5. User Events Analytics

> **Trigger**: Analytics events, tracking events, Gobbler events, event payloads, event triggers, screen tracking, event instrumentation

What Makes This Skill Unique

This skill works at **two levels**:

1. **Event Metadata** — Searches DataHub’s `userEvent` entities for event definitions, triggers, source code links, and tech owners
2. **Event Data** — Queries raw event streams in Trino for volume, trends, and user journeys

Event Metadata (DataHub)

Each event is a first-class entity in DataHub with:

- **Name**, **Description**, **Trigger** (what fires the event)
- **Source Code** — Clickable GitHub URL to the exact file
- **Tech Owner** — Git blame author
- **Screen Names** — Where the event fires
- **Category** — Business domain (Stocks, FNO, MF, Onboarding, etc.)
- **App** — Which app (Groww Android, Groww iOS, 915 Web, etc.)

Event Data (Trino)

Table	Engine	Speed	Use When
`platform_dp_pinot.default.ems_events_final`	Pinot	Fast (seconds)	Trends, top-N, dashboards (~98% accuracy for 2+ day old data)
`platform_iceberg.ems.ems_raw`	Iceberg	Slower (minutes)	Exact counts, user-level, payloads, last 1-2 days

Critical: Pinot Contains Sampled Data

Pinot stores **sampled events**. Raw `COUNT(*)` returns the sample count, not true volume. Must use:

- **Event count**: `SUM(boosting_factor * COUNT(*))` via CTE
- **User count**: Event count x `MIN(avg_user_to_record_ratio)`
- **Last 1-2 days are broken** (~1% accuracy) — always use Iceberg for recent data

App Identification

App	Filter
Groww Android	`app_package_name = 'com.nextbillion.groww' AND os_name = 'Android'`
Groww iOS	`app_package_name IS NULL AND os_name = 'iOS'`
Groww Web	`app_package_name = 'com.nextbillion.groww.web'`
915 Android	`app_package_name = 'com.groww.trade915' AND os_name = 'Android'`
915 iOS	`app_package_name = 'com.groww.trade915' AND os_name = 'iOS'`
915 Web	`app_package_name = 'com.nextbillion.915.web'`

Example Questions It Can Answer

- “Tell me about the GobblerAppLaunch event”
- “What events fire on the home screen?”
- “Who owns this event? Show me the source code”
- “How many times did event X fire yesterday?”
- “Compare event volume across Android vs iOS”
- “What’s in the payload for GobblerWidgetClick?”
- “Show me user Y’s event journey for the last hour”
- “Is event X usage growing or declining?”

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6. Onboarding Analytics

> **Trigger**: Signup, KYC, KRA, eSign, DDPI, account activation, demat, FID, onboarding conversion, TAT, bank verification, account closure, F&O activation

Tables

Table	Schema	Rows	Purpose
`onb_usercheckpoints_master`	`core_bgv`	117M	Central signup → onboarding funnel (all checkpoints)
`fact_onboarding_invest`	`core_invest`	45M	Backend processing (TE/TR/UCC/BSE/NSE) + FID dates
`fact_onboarding_fno`	`core_invest`	48M	Detailed F&O activation funnel (154 columns)
`onb_bank_fact_trino`	`core_bgv`	96.6M	Bank verification attempts
`kra_workflow_master`	`core_bgv`	29M	KRA workflow status
`onb_ddpi_master`	`core_bgv`	8.9M	DDPI activation
`esign_details`	`core_bgv`	1.5M	E-sign details (contains PII)
`account_closure_master`	`core_bgv`	180K	Account closure requests
`daily_onboarding_and_fid`	`core_invest`	1,751	Quick daily trend (pre-aggregated)

The Onboarding Funnel

```
Signup → Email Verified → Mobile Verified → PAN Entered → Bank Verified
→ Signature → E-sign → KYC Verified → Stock Onboarding (success)
                                          ↓
                                F&O Activation → Commodity Activation
```

Each step is a **timestamp column** in `onb_usercheckpoints_master`. `NULL = step not reached`.

Example Questions It Can Answer

- “Daily signup → onboarding conversion funnel”
- “Where are users dropping off in the funnel?”
- “Backend processing status (TE/TR/UCC/BSE/NSE)”
- “Bank verification success rate by mode (penny drop, UPI, reverse penny drop)”
- “KRA workflow success rate by source (CVL, DOTEX)”
- “Time-to-first-investment after onboarding”
- “F&O activation funnel with TATs”
- “DDPI activation pipeline”
- “Account closure completion rate”

Key Guardrails

- `esign_details` contains PII (PAN, Aadhaar, bank numbers) — never SELECT *

- DO NOT USE `invest_user_fid_info` — stale since Jun 2024
- DO NOT USE `daily_onb_conversions` — EMPTY, pipeline broken
- `fact_onboarding_fno` has 154 columns — select only needed funnel steps

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7. UPI Payments & Transactions Analytics

> **Trigger**: UPI payments, autopay, mandates, payout, bill payments, UPI onboarding, transaction success rates

Tables

Table	Schema	Rows	Status
`groww_upi_onb_user_fact`	`core_bgv`	66.9M	FRESH — UPI onboarding users
`groww_upi_mandate_fact_table`	`core_bgv`	27.2M	FRESH — Mandate lifecycle
`elements_upiautopay_recon_fact_v2_trino_v1`	`core_bgv`	1.6M	FRESH — Rolling 7d autopay recon
`elements_payout_fact_ops_dc`	`core_bgv`	729K	FRESH — Rolling 4d payouts
`gupi_bill_order_fact`	`core_bgv`	19.8M	STALE (Apr 2025) — Bill payments
`gupi_daily_user_session_txn_v2`	`core_bgv`	572M	STALE (Oct 2024) — Daily sessions

UPI Onboarding Funnel

```
Registration → Device Bound → Bank Linked → VPA Created → Fully Activated
```

Mandate Status Distribution

Status	%	Description
`REVOKED`	46%	User/system revoked
`REQUEST_EXPIRED`	16%	Approval window expired
`ACTIVE`	13%	Currently active
`COMPLETED`	9%	Fully executed
`FAILURE`	8%	Creation/approval failed

Example Questions It Can Answer

- “UPI onboarding funnel and activation rate”
- “Autopay reconciliation success rate by provider (HDFC, ICICI)”
- “Mandate lifecycle distribution”

- “Payout volume by method (IMPS, NEFT, UPI)”
- “Bill payment performance and reward analysis”
- “Debit transaction success rate by flow”

Key Guardrails

- Most timestamps are **epoch milliseconds** — use `from_unixtime(col / 1000)`
- Rolling window tables (autopay recon, payouts) contain only recent data — NOT for historical trends
- Some tables are stale (debit: Feb 2024, sessions: Oct 2024) — check freshness first

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8. Engagement & Communications Analytics

> **Trigger**: DAU, sessions, push notifications, email, SMS, WhatsApp, DND, app install, campaign performance

Tables

Table	Schema	Rows/Day	Channel
`engagement_pn_narad_master`	`core_bgv`	170-250M	Push notifications (Narad)
`engagement_email_backend_master`	`core_bgv`	3-6M	Email (Backend)
`engagement_sms_backend_master`	`core_bgv`	500-650K	SMS
`engagement_whatsapp_backend_master`	`core_bgv`	100-140K	WhatsApp
`engagement_user_session_daily`	`core_bgv`	8M	DAU (aggregated)
`engagement_dau_indepth`	`core_bgv`	8M	App engagement metrics
`engagement_session_indepth`	`core_bgv`	**65M**	Per-session detail (**57B total**)
`engagement_app_fact`	`core_bgv`	Weekly	App install/reachability
`engagement_dnd_fact`	`core_bgv`	70K	DND state changes

Push Notification Funnel (Typical Day)

```
Total: ~3M → Sent: ~2M (66%) → Received: ~440K (22%) → Shown: ~438K → Clicked: ~19K (4.3% CTR)
```

PN Status Distribution

Status	%
`SENT`	72%
`OPTED_OUT`	9%
`VENDOR_FAILURE`	9%

`FREQUENCY_CAPPED`	8.6%
`EXPIRED`	1.7%

Example Questions It Can Answer

- “DAU trend — app vs web vs transacting users”
- “Push notification delivery funnel and CTR”
- “Email open/click rates by event group”
- “SMS delivery rate by vendor (Infobip, Vconnect, Airtel)”
- “WhatsApp read receipt rates”
- “Cross-channel volume comparison”
- “DND opt-out trends by OS”
- “Campaign performance — top campaigns by CTR”
- “Session quality — bounce rate and avg duration”
- “App install and push reachability rates”

Key Guardrails

- `engagement_session_indepth` has **57 BILLION rows** — SINGLE DAY queries only
- PN Narad `os` column has mixed casing (`android` vs `ANDROID`) — always use `LOWER(os)`
- DND `s7` column is VARCHAR (`'true'`/`'false'`), not boolean
- Prefer `engagement_user_session_daily` for DAU over `engagement_session_indepth`

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9. MTF & Pledge Analytics

> **Trigger**: MTF orders, margin trading, loan book, shortfall, market share, pledge, unpledge, collateral, OCO orders

Tables

Table	Schema	Rows	Purpose
`aay_mtf_fid_agg`	`core_invest`	180K	Daily MTF turnover by stock (FRESH)
`aay_mtf_book_agg`	`core_invest`	243	Daily book snapshot (FRESH)
`aay_mtf_market_share_data`	`core_invest`	367K	Groww vs market by stock (FRESH)
`aay_mtf_shortfall`	`core_invest`	72K	Shortfall/margin call analysis (FRESH)
`aay_mtf_oco_base_slippage`	`core_invest`	83K	OCO order slippage
`mtf_transactions_v2`	`core_invest`	2.0M	Raw MTF orders (stale Nov 2024)
`fact_pledge_transaction`	`core_invest`	50M	Pledge transactions
`fact_user_date_isin_pledge_collateral`	`core_invest`	217M	Daily collateral by user x ISIN

`fact_unpledge_request`	`core_invest`	4.2M	Unpledge requests

Example Questions It Can Answer

- “MTF book size trend”
- “Top MTF stocks by turnover”
- “Groww’s MTF market share by stock”
- “MTF shortfall analysis”
- “OCO order slippage — trigger price vs fill price”
- “Daily pledge/unpledge volume”
- “Top pledged stocks by collateral value”
- “Unpledge success rate by source (user vs risk invocation)”

Key Guardrails

- Prefer `aay_mtf_fid_agg` (fresh daily) over `mtf_transactions_v2` (stale Nov 2024)
- `aay_mtf_market_share_data` values are already in crores — no conversion needed
- `fact_user_date_isin_pledge_collateral` is 217M rows — always filter on `transaction_date`

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10. Commodities Trading Analytics

> **Trigger**: Commodity trading, MCX orders, gold/silver/crude oil, commodity futures/options, commodity PnL

Tables

Table	Schema	Rows	Purpose
`commodity_day_view_seg`	`core_invest`	684K	Pre-aggregated daily metrics (preferred)
`commodity_turnover_detailed`	`core_invest`	159K	Turnover by contract and asset
`commodity_order_master`	`core_invest`	71.6M	Raw commodity orders
`commodities_pnl_master_v3`	`core_invest`	27.6M	Position-level PnL
`commodity_overview2`	`core_invest`	1	Single-row platform summary
`commodity_leaderboard`	`core_invest`	302K	User lifetime rankings

Top Underlyings

`GOLD`, `GOLDM`, `SILVER`, `SILVERM`, `SILVERMIC`, `CRUDEOIL`, `CRUDEOILM`, `NATURALGAS`, `NATGASMINI`, `COPPER`, `ZINC`

Example Questions It Can Answer

- “Daily commodity order volume and success rate”
- “Futures vs Options breakdown”

- “Top commodities by turnover (gold, silver, crude)”
- “Platform breakdown (Android, iOS, web)”
- “Commodity PnL distribution”
- “Compliance-flagged order analysis”
- “Holding time analysis (scalping vs intraday)”

Key Guardrails

- Prefer `commodity_day_view_seg` (684K rows, pre-aggregated) over `commodity_order_master` (71.6M, no partition)
- Timestamps in order/PnL tables are epoch milliseconds — use `from_unixtime(col / 1000)`
- All commodity trading is on MCX exchange
- Underlying naming varies: `GOLD` vs `GOLDM` (mini) — use `asset` column in turnover tables for clean grouping

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11. Help & Support Analytics

> **Trigger**: Support tickets, resolution times, CSAT, chatbot deflection, voice calls, IVR, agent quality

Tables

Table	Schema	Rows	Status
`hns_helpdeskticketts_master_trino`	`core_bgv`	12M	ACTIVE — Primary ticket table
`hns_chatbot_fact_table_v3`	`core_bgv`	135K	ACTIVE — Rolling 3d chatbot sessions
`hns_voicecallrequests_master_trino`	`core_bgv`	4.4M	ACTIVE — Call-me-back requests
`agent_call_insights_analytics`	`core_bgv`	124	ACTIVE — Agent quality scoring (new)
`hns_exotel_samples`	`core_bgv`	80	ACTIVE — IVR call samples
`hns_csat_dashboard`	`core_bgv`	216K	STALE (Mar 2024)
`hns_gpt_fact`	`core_bgv`	8.5M	STALE (Jun 2024)

Channel Source Codes (Helpdesk)

Code	Channel
1	Email
2	Portal
3	Chat
7	Feedback Widget
10	Outbound

Example Questions It Can Answer

- “Daily ticket volume by product (Stocks, MF, F&O)”
- “Average resolution time by product”
- “Chatbot self-service rate and GPT effectiveness”
- “Voice call connect rate by product”
- “Agent quality scorecard”
- “Channel mix trend (chat vs email vs portal)”
- “IVR call path analysis”
- “Top issue categories”

Key Guardrails

- `hns_freshchat_tickets_master_trino` is BROKEN — do not use
- Chatbot table is a rolling 3-day window — no historical data
- Exotel is a balanced sample (20 per category) — not for volume calculations
- Time fields (`resolution_time`, `first_response_time`) are in seconds, not milliseconds

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12. 915 Trading Analytics

> **Trigger**: 915 platform, 915 DAU/MTU, 915 PnL, 915 sessions, 915 commodity, 915 onboarding, 915 user segments

915 is Groww’s separate trading app focused on F&O and derivatives, distinct from the main Groww app.

Tables

Table	Schema	Rows	Purpose
`userday_details_915`	`core_invest`	1.9M	Daily activity (sessions + orders + commodity)
`usersession_915`	`core_invest`	1.9M	Session gap/retention analysis
`usertxn_915`	`core_invest`	285K	Daily transaction details
`onb_915_master`	`core_bgv`	44K	Onboarding funnel
`mis_central_915_ab`	`core_bgv`	17	Monthly KPIs (17 rows!)
`user_cohorts_915`	`dump_invest`	5.5M	Trading behavior metrics
`user_cohorts_915_buckets`	`dump_invest`	5.5M	Pre-computed segments

Pre-Computed User Segments

Segment	Values
Scalper	`non_scalper`, `scalper_1h`, `scalper_5min`

Buyer/Seller	`buyer_only`, `buyer_seller`, `seller_only`
Profitability	`profit_making`, `loss_making`
Platform	`mobile_centric`, `web_centric`, `mixed`
Recency	`recent_trader`, `inactive_last_month`
Frequency	`frequent`, `occasional`, `rare`

Example Questions It Can Answer

- “915 DAU/MTU/DTU trends”
- “Order success rate by platform (web/iOS/Android)”
- “Session retention — D7 return rate”
- “Monthly KPIs — signups, MTU, turnover, NPS”
- “915 onboarding funnel and conversion”
- “User PnL distribution”
- “Scalper vs non-scalper analysis”
- “Commodity trading adoption on 915”
- “Platform preference (web vs mobile)”

Key Guardrails

- 915 data starts from June/July 2025 — no data before this
- `user_cohorts_915` has 5.5M rows but 915 ATU is only ~26K — cohort tables include broader F&O users. Inner join with `onb_915_master` for 915-only
- `mis_central_915_ab` has duplicate rows — use `WHERE refresh_month = month` for latest

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13. Other Skills

Context Push Analytics

Contextual push notification analytics — trigger patterns, delivery, and user response for context-aware notifications.

User Financial Health Analytics

User financial wellness metrics — AUM health, investment diversification, and financial behavior patterns.

Datahub Skills

Meta-skill for navigating DataHub itself — finding datasets, understanding lineage, managing metadata.

Cross-Skill Reference

Universal Join Key

All user-level tables across ALL skills use **`user_account_id`** (format: `ACC<digits>`) as the primary join key.

Schema Directory

Schema	Tables	Domain
`platform_iceberg.core_invest`	Stocks, F&O, MF, MTF, Commodity, 915	Trading & Investment
`platform_iceberg.core_bgv`	Growth, Onboarding, UPI, Engagement, H&S	User Lifecycle & Operations
`platform_iceberg.dwh_invest`	MF (enriched)	Mutual Funds Data Warehouse
`platform_iceberg.dump_invest`	915 cohorts	Periodic Snapshots
`platform_iceberg.ems`	Event data	Raw Events (Iceberg)
`platform_dp_pinot.default`	Event data	Sampled Events (Pinot)

Table Size Reference (Largest Tables)

Table	Rows	Skill	Must Filter?
`user_product_txn_master_fact`	8.3B	User Master	`order_date`
`fact_transaction_stocks`	7.1B	Stocks	`order_date`
`fno_order_master`	3.8B	F&O	`pt`
`core_stocks_order`	2.5B	Stocks	`order_date`
`fno_pnl_master`	1.9B	F&O	`pt`
`engagement_session_indepth`	57B total	Engagement	`session_date` (1 day only!)
`engagement_pn_narad_master`	250M/day	Engagement	`event_date`
`growth_user_master_ultimate`	117M	User Master	Recommended
`onb_usercheckpoints_master`	117M	Onboarding	`groww_signup_time`

Stale Table Warnings

Table	Last Data	Skill	Use Instead
`stocks_order`	Oct 2025	Stocks	`core_stocks_order`
`invest_user_fid_info`	Jun 2024	Onboarding	`fact_onboarding_invest` FID columns
`daily_onb_conversions`	EMPTY	Onboarding	`daily_onboarding_and_fid`

`fno_usersegmentation_monthly`	Oct 2025	F&O	Use order tables for current data
`groww_upi_debit_fact_table`	Feb 2024	UPI	Historical only
`hns_freshchat_tickets_master_trino`	BROKEN	H&S	Do not use
`hns_csat_dashboard`	Mar 2024	H&S	Historical CSAT only
`mtf_transactions_v2`	Nov 2024	MTF	`aay_mtf_fid_agg`

How to Use a Skill

Skills are loaded automatically by the AI assistant when your question matches the skill's domain. You can also reference them explicitly:

```
"Use the Stocks skill to answer: what was yesterday's order volume?"
"Load the F&O analytics skill and show me power trader stats"
"Using the MF skill, what's the SIP creation trend this month?"
```

No configuration needed — just ask your question and Compass handles the rest.